

Giannis Karamanolakis

Senior Applied Scientist, Amazon AGI Labs
Ph.D. in Computer Science, Columbia University
Scholar: 20+ papers, 2000+ citations, h-index 10+

WWW: gkaramanolakis.github.io
E-mail: giannis.karamanolakis@gmail.com
Tel: +1 (917)-331-8844

Senior scientist and founding member of the Amazon Nova team, contributing to the development of foundation models from early design to production by leading projects across **pre-training, post-training, data pipelines, and evaluation**. **10+ years** of research on efficient frameworks for **teaching models new capabilities from limited or noisy data**.

Professional and Research Experience

5. **Amazon**, New York, USA. 09/2022 - now
Senior Applied Scientist, Amazon AGI Labs, Pre-training Foundations horizontal team 2023-now
Applied Scientist, Amazon Alexa AI, NLU Modeling team 2022-2023
 - Co-developed the top-ranked internal language model, informing the decision to scale and launch Amazon Nova.
 - Owned large-scale production training jobs and led cross-functional investigations to unblock critical launches.
 - Led end-to-end technical initiatives spanning pre-training, post-training, data pipelines, and evaluation:
 1. **Tool Calling**: training & eval methods for the first LLM-powered Alexa+ release in 2023 (Blog, WSJ, CNET).
 2. **Evals**: the evaluation strategy that enabled fast data and training decisions, securing top rank internally (2023).
 3. **Data Quality Gating**: the protocol being used by the AGI organization to greenlight text data submissions.
 4. **Data Filtering and Cleaning**: techniques improving text data quality and boosting token efficiency.
 5. **Factuality**: training & eval methods for stronger memorization of long-tail factual knowledge (e.g., SimpleQA).
 6. **Agentic Long Context**: the context extension recipe improving Agentic & OpenAI-MRCR performance.
 7. **Data Mixing**: recipes for automatic data ratio optimization, improving end-to-end accuracy and efficiency.
 8. **Continual Pre-training**: recipes boosting accuracy for key domains, enabling model customization (Forge Blog).
 9. **Agentic Self-Improvement**: methods for automatic scaffold optimization toward stronger AI R&D agents.
 - Steered direction for 10+ scientists toward production deployments and publications (e.g., LoRA, model merging).
4. **Columbia University**, Department of Computer Science, New York, USA. 09/2017 - 08/2022
Graduate Research Assistant, Advisors: Luis Gravano, Daniel Hsu, Groups: ML, NLP, Database Research.
 - Designed frameworks for training neural networks with limited or noisy supervision (e.g., EMNLP'19/20, TACL'24).
 - Deployed AI systems in health departments (NYC, LA) for public health (NSF, New York Times, Washington Post).
 - Co-developed an early large-scale instruction-tuning dataset for LLM training (EMNLP '22; 800+ citations).
3. **Microsoft Research AI**, Redmond, USA. 05/2020 - 08/2020
Research Scientist Intern, Supervisors: Ahmed Awadallah, Subho Mukherjee, Guoqing Zheng (LTI team).
 - Designed a semi-supervised learning method (NAACL '21) and a benchmark (NAACL '22) for weak supervision.
2. **Amazon**, Seattle, USA. 05/2019 - 12/2019
Applied Scientist / Machine Learning Scientist Intern, Supervisor: Xin Luna Dong (Product Graph team).
 - Developed a knowledge extraction method (ACL '20) with +15% coverage in Amazon's knowledge graph (KDD '20).
1. **Behavioral Signals**, Los Angeles, USA (remote from Athens, Greece). 03/2016 - 06/2017
Machine Learning Engineer, Supervisors: Shri Narayanan, Alexandros Potamianos.
 - Trained and deployed statistical models for recognizing core emotions from conversational speech.

Education

2. **Columbia University**, New York, NY, United States. 09/2017 - 08/2022
Ph.D. in Computer Science, M.Phil. in Computer Science, M.Sc. in Computer Science GPA: 4.26/4.00
 Advisors: Prof. Luis Gravano, Prof. Daniel Hsu.
 Ph.D. Dissertation: "Efficient Machine Teaching Frameworks for Natural Language Processing".
1. **National Technical University of Athens**, Athens, Greece. 09/2011 - 07/2017
M.Eng. in Electrical & Computer Engineering (5-year study program, M.Sc. equivalent) GPA: 9.26/10.00
 Advisor: Prof. Alexandros Potamianos.
 Dissertation: "Grounding Word Semantic Representations to Acoustic and Visual Perceptual Modalities".

Selected Publications

For all publications, please see Google Scholar (20+ papers, 2000+ citations, h-index 10+)

1. Amazon Artificial General Intelligence. **Amazon Nova 2: Multimodal reasoning and generation models.** *Amazon Technical Reports*, 2025.
2. Amazon Artificial General Intelligence. **The Amazon Nova family of models: Technical report and model card.** *Amazon Technical Reports*, 2024.
3. **Giannis Karamanolakis**, Daniel Hsu, and Luis Gravano. **Interactive Machine Teaching by Labeling Rules and Instances.** *TACL*, 2024 and *EMNLP*, 2025.
4. Yizhong Wang, Swaroop Mishra, et al. **Super-NaturalInstructions: Benchmarking Generalization via Declarative Instructions on 1600+ NLP Tasks.** In *EMNLP*, 2022.
5. **Giannis Karamanolakis**, Subhabrata Mukherjee, Guoqing Zheng, and Ahmed Hassan Awadallah. **Self-Training with Weak Supervision.** In *NAACL*, 2021 (Oral).
6. **Giannis Karamanolakis**, Jun Ma, and Xin Luna Dong. **TXtract: Taxonomy-Aware Knowledge Extraction for Thousands of Product Categories.** In *ACL*, 2020 (Oral).
7. **Giannis Karamanolakis**, Daniel Hsu, and Luis Gravano. **Leveraging Just a Few Keywords for Fine-Grained Aspect Detection Through Weakly Supervised Co-Training.** In *EMNLP*, 2019 (Oral).

Selected Fellowships, Honors, and Awards

1. **Columbia Ph.D. Award**, Davide Giri Memorial Prize, US 05/2022
2. **Scholarships from Leventis & Gerontelis Foundations**, US 06/2018, 08/2019, 08/2020, 08/2021
3. **Joseph F. Traub Fellowship**, Columbia University, US 04/2018
4. **Addressing the Opioid Epidemic, Design Challenge Award**, Columbia University, US 12/2017
5. **Columbia Engineering Presidential Fellowship**, US 08/2017 - 05/2022

Selected Skills, Talks and Academic Service

- **ML & Systems:** Python, C/C++, Bash, , SQL, PySpark, NeMo Megatron, DeepSpeed, HF, vLLM, Kubernetes, EC2.
- **Business:** Agile development (Scrum, Kanban), Business Foundation Specialization (UPenn Wharton School).
- **Invited Industry Talks:** Microsoft Research ('22), Two Sigma ('20), RelationalAI ('20), NYAS ('20), AT&T Labs ('19).
- **Conference Talks:** EMNLP '25, NAACL '21/'22, DeeLIO '20, ACL '20, EMNLP '19, RecSys '18, InterSpeech '16.
- **Reviewer Activity:** Regular reviewer at NeurIPS (top 10% reviewer), ICML (expert reviewer), ICLR, ARR, etc.
- **Mentoring:** Designed 20+ research projects for students and scientists (10+ published papers + prod deployments).

All Publications

24. Amazon Artificial General Intelligence. **Amazon Nova 2: Multimodal reasoning and generation models.** *Amazon Technical Reports*, 2025.
23. Yuhang Zhu, **Giannis Karamanolakis**, Victor Soto, Mayank Kulkarni, Anna Rumshisky, Furong Huang, Wei Ai, and Jianhua Lu. **MergeME: Model Merging Techniques for Homogeneous and Heterogeneous MoEs.** *NAACL*, 2025.
22. **Giannis Karamanolakis**, Daniel Hsu, and Luis Gravano. **Interactive Machine Teaching by Labeling Rules and Instances.** *EMNLP*, 2025 (accepted via TACL).
21. Amazon Artificial General Intelligence. **Amazon Nova Premier: Technical report and model card .** *Amazon Technical Reports*, 2025.
20. Amazon Artificial General Intelligence. **The Amazon Nova family of models: Technical report and model card.** *Amazon Technical Reports*, 2024.
19. Namrata Shivagunde, Mayank Kulkarni, **Giannis Karamanolakis**, Jack GM FitzGerald, Yannick Versley, Saleh Soltan, Volkan Cevher, Jianhua Lu, and Anna Rumshisky. **Approximations may be all you need: Towards pre-training LLMs with low-rank decomposition and optimizers.** *NeurIPS 2024 Workshop on Efficient Natural Language and Speech Processing (ENLSP-IV)*, 2024.
18. **Giannis Karamanolakis**, Daniel Hsu, and Luis Gravano. **Interactive Machine Teaching by Labeling Rules and Instances.** *Transactions of the Association for Computational Linguistics (TACL)*, 2024.
17. Guoqing Zheng, **Giannis Karamanolakis**, Kai Shu, and Ahmed Hassan Awadallah. **WALNUT: A Benchmark on Semi-weakly Supervised Learning for Natural Language Understanding.** In *NAACL*, 2022 (Oral).
16. Yizhong Wang, Swaroop Mishra, Pegah Alipoormolabashi, Yeganeh Kordi, Amirreza Mirzaei, Anjana Arunkumar, Arjun Ashok, Arut Selvan Dhanasekaran, Atharva Naik, David Stap, Eshaan Pathak, **Giannis Karamanolakis**, Haizhi Gary Lai, Ishan Purohit, Ishani Mondal, Jacob Anderson, Kirby Kuznia, Krima Doshi, Maitreya Patel, Kuntal Kumar Pal, Mehrad Moradshahi, Mihir Parmar, Mirali Purohit, Neeraj Varshney, Phani Rohitha Kaza, Pulkit Verma, Ravsehaj Singh Puri, Rushang Karia, Shailaja Keyur Sampat, Savan Doshi, Siddhartha Mishra, Sujana Reddy, Sumanta Patro, Tanay Dixit, Xudong Shen, Chitta Baral, Yejin Choi, Hannaneh Hajishirzi, Noah A. Smith, and Daniel Khashabi. **Super-NaturalInstructions: Benchmarking Generalization via Declarative Instructions on 1600+ NLP Tasks .** In *EMNLP*, 2022.
15. **Giannis Karamanolakis**, Subhabrata Mukherjee, Guoqing Zheng, and Ahmed Hassan Awadallah. **Self-Training with Weak Supervision.** In *NAACL*, 2021 (Oral).
14. Ivy Cao, Zizhou Liu, **Giannis Karamanolakis**, Daniel Hsu, and Luis Gravano. **Quantifying the Effects of COVID-19 on Restaurant Reviews.** In *EMNLP 9th International Workshop on Natural Language Processing for Social Media (SocialNLP)*, 2021 (Oral).
13. Tejit Pabari, **Giannis Karamanolakis**, Max Mauerman, Beth Tellman, Mitchell Thomas, and Eugene Wu. **Flood Event Extraction from Newspaper Media in Bangladesh.** In *Book of Abstracts of the 8th International Conference on Water and Flood Management (ICWFM)*, 2021.
12. **Giannis Karamanolakis**, Daniel Hsu, and Luis Gravano. **Cross-Lingual Text Classification with Minimal Resources By Transferring a Sparse Teacher.** In *Findings of EMNLP*, 2020.
11. Ziyi Liu, **Giannis Karamanolakis**, Daniel Hsu, and Luis Gravano. **Detecting Foodborne Illness Complaints in Multiple Languages Using English Annotations Only.** In *EMNLP Workshop on Health Text Mining and Information Analysis*, 2020 (Oral).
10. **Giannis Karamanolakis**, Jun Ma, and Xin Luna Dong. **TXtract: Taxonomy-Aware Knowledge Extraction for Thousands of Product Categories.** In *ACL*, 2020 (Oral).

9. Xin Luna Dong, Xiang He, Andrey Kan, Xian Li, Yan Liang, Jun Ma, Yifan Ethan Xu, Chenwei Zhang, Tong Zhao, Gabriel Blanco Saldana, Saurabh Deshpande, Alexandre Michetti Manduca, Jay Ren, Surender Pal Singh, Fan Xiao, Haw-Shiuan Chang, **Giannis Karamanolakis**, Yuning Mao, Yaqing Wang, Christos Faloutsos, Andrew McCallum, and Jiawei Han. **AutoKnow: Self-Driving Knowledge Collection for Products of Thousands of Types**. In *KDD*, 2020 (Oral).
8. **Giannis Karamanolakis**, Daniel Hsu, and Luis Gravano. **Leveraging Just a Few Keywords for Fine-Grained Aspect Detection Through Weakly Supervised Co-Training**. In *EMNLP*, 2019 (Oral).
7. **Giannis Karamanolakis**, Daniel Hsu, and Luis Gravano. **Weakly Supervised Attention Networks for Fine-Grained Opinion Mining and Public Health**. In *EMNLP Workshop on Noisy User-generated Text (WNUT)*, 2019 (Oral).
6. **Giannis Karamanolakis**, Daniel Hsu, and Luis Gravano. **Training Neural Networks for Aspect Extraction Using Descriptive Keywords Only**. In *ICLR 2nd Workshop on Learning from Limited Labeled Data (LLD)*, 2019.
5. Katy Ilonka Gero, **Giannis Karamanolakis**, and Lydia Chilton. **Transfer Learning for Style-Specific Text Generation**. In *NeurIPS Workshop on Machine Learning for Creativity and Design*, 2018.
4. **Giannis Karamanolakis**, Kevin Raji Cherian, Ananth Ravi Narayan, Jie Yuan, Da Tang, and Tony Jebara. **Item Recommendation with Variational Autoencoders and Heterogenous Priors**. In *RecSys Workshop on Deep Learning for Recommender Systems*, 2018 (Oral).
3. Georgios Paraskevopoulos, **Giannis Karamanolakis**, Elias Iosif, Aggelos Pikrakis, and Alexandros Potamianos. **Sensory-Aware Multimodal Fusion for Word Semantic Similarity Estimation**. In *EUSIPCO Multimodal Processing, Modeling, and Learning (Multi-Learn) Workshop*, 2017 (Oral).
2. **Giannis Karamanolakis**, Elias Iosif, Athanasia Zlatintsi, Aggelos Pikrakis, and Alexandros Potamianos. **Audio-based Distributional Semantic Models for Music Auto-tagging and Similarity Measurement**. In *EUSIPCO Multimodal Processing, Modeling, and Learning Workshop*, 2017 (Oral).
1. **Giannis Karamanolakis**, Elias Iosif, Athanasia Zlatintsi, Aggelos Pikrakis, and Alexandros Potamianos. **Audio-Based Distributional Representations of Meaning Using a Fusion of Feature Encodings**. In *INTERSPEECH*, 2016 (Oral).